



The View from Edge-on

Hone your observing skills on superthin galaxies.

WHEN THE PLANE of a spiral galaxy tilts directly toward our line of sight, the edge-on perspective can provide a breathtaking view of the bright central bulge and equatorial dust lane. NGC 4565 epitomizes the beauty and symmetry of these features. But not all disk galaxies contain large bulges and obscuring dust bands.

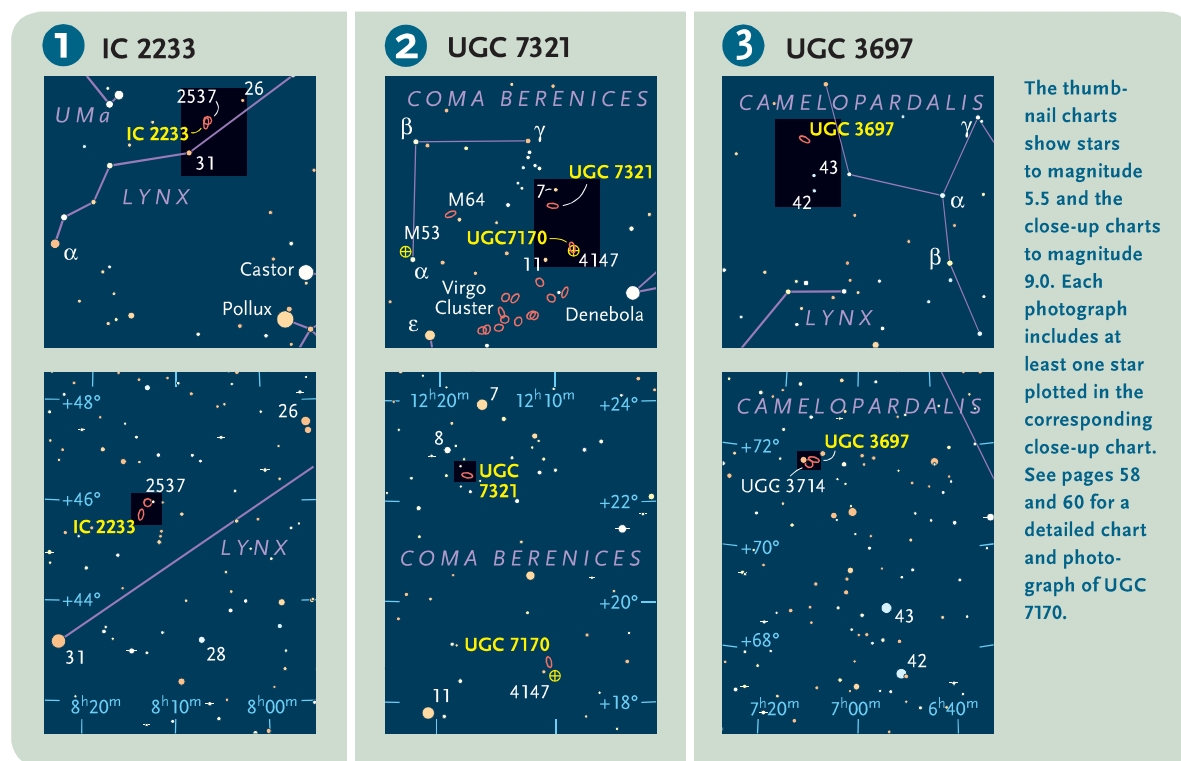
Highly flattened edge-on galaxies with little or no bulge are called flat galaxies. The *Revised Flat Galaxy Catalogue* (RFGC) includes 4,236 galaxies with an axial ratio of 7:1 or higher and a major axis extending at least 0.6'.

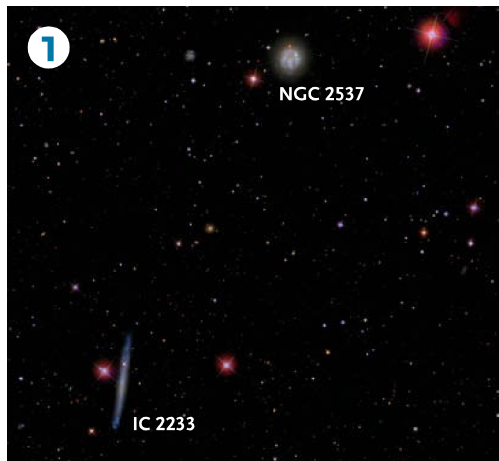
The most extreme edge-ons are known as superthin galaxies, a term coined in 1981 by researchers Jean Goad of Kitt Peak and Morton Roberts of the National Radio Astronomy Observatory. These wafer-thin galaxies have axial ratios ranging from 9:1 to 20:1 and consist of gas-rich late-type spirals (Sc, Sd, or Sm) that lack a discern-

ible bulge or organized dust lane. Generally, superthins have a very low, even surface brightness, so they're much tougher visual targets than their magnitudes suggest. Follow along and we'll explore four of these ethereal wisps.

One of my favorites is **IC 2233**, also known as RFGC 1340 (*S&T*: March 2010, page 64). This 12.6-magnitude superthin is located 17' southeast of the peculiar Bear-Paw Galaxy (NGC 2537) in Lynx. In my 18-inch Dobsonian at 100×, a needle-thin slash is just visible, tilting north-northwest to south-southeast. Use higher magnification to darken the background and boost the contrast. This sliver is easier to view at 220× and extends 1.8' × 0.2'. A 14th-magnitude star is perched at the north tip, and a pair of 10th- and 13th-magnitude stars lies just off the east side.

Last April I had a stunning view of this superthin at 510× in the 48-inch f/4.0 Dobsonian of Texan amateur





Superthin Galaxies

Galaxy	Const.	Mag(v)	Size	RA	Dec.
IC 2233	Lyn	12.6	5.2' × 0.6'	8 ^h 14.0 ^m	+45° 45'
UGC 7321	Com	13.4	5.5' × 0.3'	12 ^h 17.6 ^m	+22° 32'
UGC 7170	Com	14.3	3.3' × 0.3'	12 ^h 10.6 ^m	+18° 50'
UGC 3697	Cam	12.9	2.2' × 0.2'	7 ^h 11.3 ^m	+71° 50'

Angular sizes are from recent catalogs. Visually, an object's size is often smaller than the cataloged value and varies according to the aperture and magnification of the viewing instrument. Right ascension and declination are for equinox 2000.0.

Jimi Lowrey. With this remarkable instrument, IC 2233 stretched a full 4' × 0.25' and displayed a mottled texture on the north side of the 14th-magnitude star.

Next let's head 4° southwest of the Coma Berenices Star Cluster to 13.4-magnitude **UGC 7321**. With a highly flattened disk yielding an axial ratio of more than 18:1, UGC 7321 is the prototype superthin galaxy. Careful viewing at 220× teases out a ghostly filament spanning 4.0' × 0.25' west-southwest to east-northeast. A 14th-magnitude star lies just beyond the eastern tip.

While in Coma Berenices, drop 4° south-southwest to the globular cluster NGC 4147. Little-known **UGC 7170** is a challenging 14.3-magnitude superthin situated 18' north-northeast in the same low-power field, as shown on page 60. I usually first notice the slightly brighter central region at 100×. Using averted vision at 220×, I can glimpse razor-thin extensions reaching 1.5' × 0.5' and sloping south-southwest to north-northeast. Images reveal a very subtle bend to the disk. I wonder if this feature can be detected visually?

As a math teacher, I was fascinated when I first saw an image of **UGC 3697**, also known as the Integral Sign Galaxy. This distorted superthin has a striking S-shaped warp with curved extensions at the tips simulating a skinny version of the familiar calculus symbol. The twist

in UGC 3697's disk is probably the result of a tidal interaction with a nearby neighbor.

UGC 3697 is located off the beaten path in Camelopardalis, just 7.6' northwest of 11.9-magnitude UGC 3714, one of the brightest galaxies missing from the NGC and IC. The Integral Sign appears as a very faint, exceedingly thin streak, roughly 2.2' × 0.2', piercing the sky in an east-west direction. I was disappointed, though, when I failed to detect the intriguing curved tips with my 18-inch.

But it's a different story in Lowrey's 48-inch behemoth. At 510× UGC 3697 spans 3.3' from tip to tip with a maximum thickness of 10" (roughly a 20:1 axial ratio!). Near the west end the galaxy clearly hooks to the northwest and fades, terminating at an elongated knot. The east end exhibits a more gradual, subtle bending. Two faint stars straddle the galaxy, just off the north and southwest side.

If I've whetted your appetite for observing flat or superthin galaxies, check out the Astronomical League's "Flat Galaxies Club" at www.astroleague.org/observing. Its list features 222 galaxies from the RFGC, with pins and certificates awarded for reaching observing milestones. ♦

Steve Gottlieb loves galaxies bright and faint, fat and thin, and everything in between. To discuss them, send e-mail to steve_gottlieb@comcast.net.

